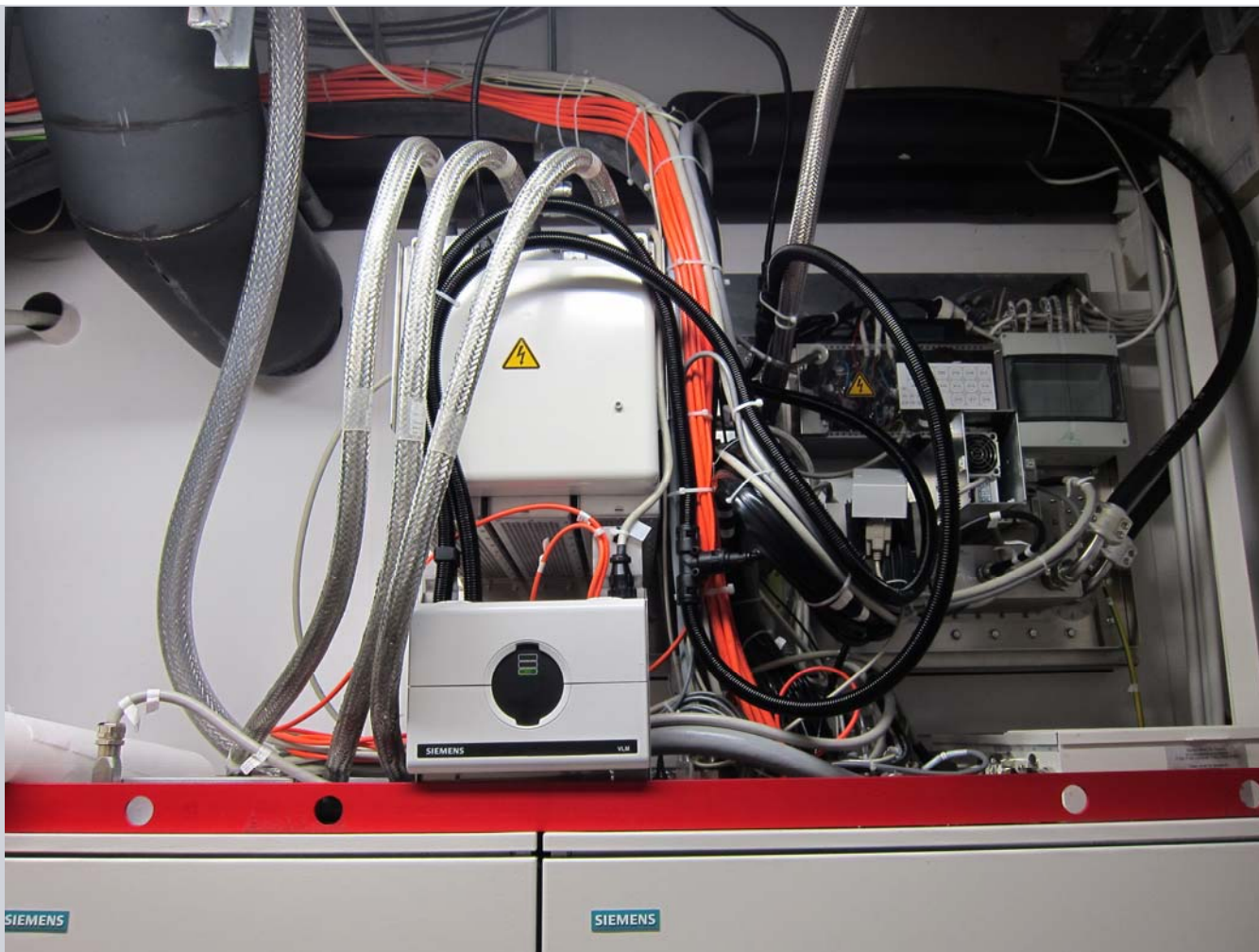


Trouble Shooting – Gradient Supervision

SIEMENS



- 1. Short Introduction to VLM-function**
- 2. Currently seen problems**
- 3. Trouble shooting flow chart**
- 4. Installation of an additional T-piece in the equipment**



The VLM-device has only 3 status LEDs:

- OK (default)
- FAULT (flashing, or static – depending on the fault severity – results in a customer pop-up)
- FIRE (turns off the GPA modulator immediately and shuts down the system 1 min. Later)

The RESET-button is used for resetting FAULTs or normalising the device



Currently seen problems – differential pressure related problems

The major problem currently seen in the field is the FAULT LED flashing or being on permanently.

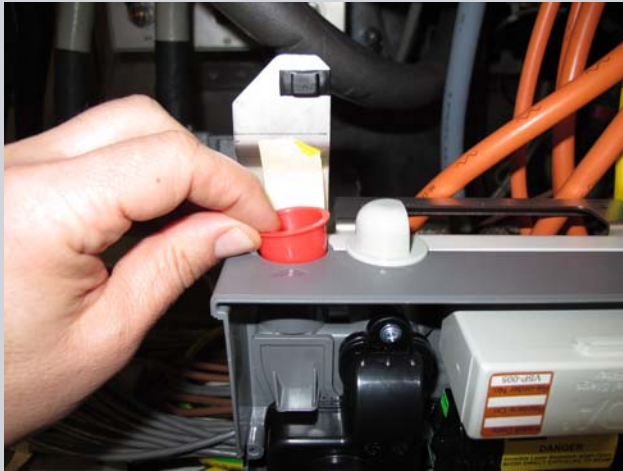
This problem is mainly caused by an incorrect air-flow (Required air-flow range between 12 and 24 ltrs./min. for normalisation and a maximum deviation of +/- 20% from normalised value).

Possible reasons for air-flow problems:

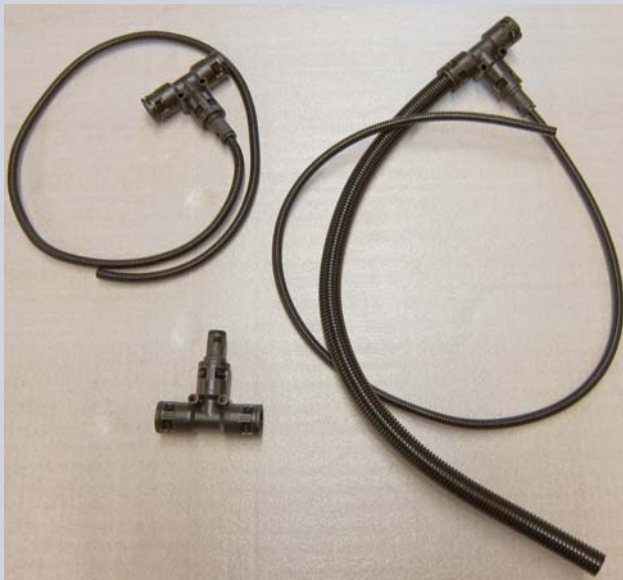


1. Excessive differential pressure between RF-cabin and equipment-room (>80 Pa). This should be below 40 Pa. This kind of problem can be typically verified by:
 - RF door closed (VLM normalised and status „OK“)
 - RF door open (VLM status „FAULT“)

Currently seen problems – Installation problems



2. Blind cap not removed from VLM-input.
This will result in zero flow and a FAULT condition.



3. Leaking hose connections.
(make sure the hose is properly cut in the groove ground and does have smooth edges. A properly sealed hose has noticeable friction, while you try to turn it in the coupling)

For internal use only

Currently seen problems – Installation problems

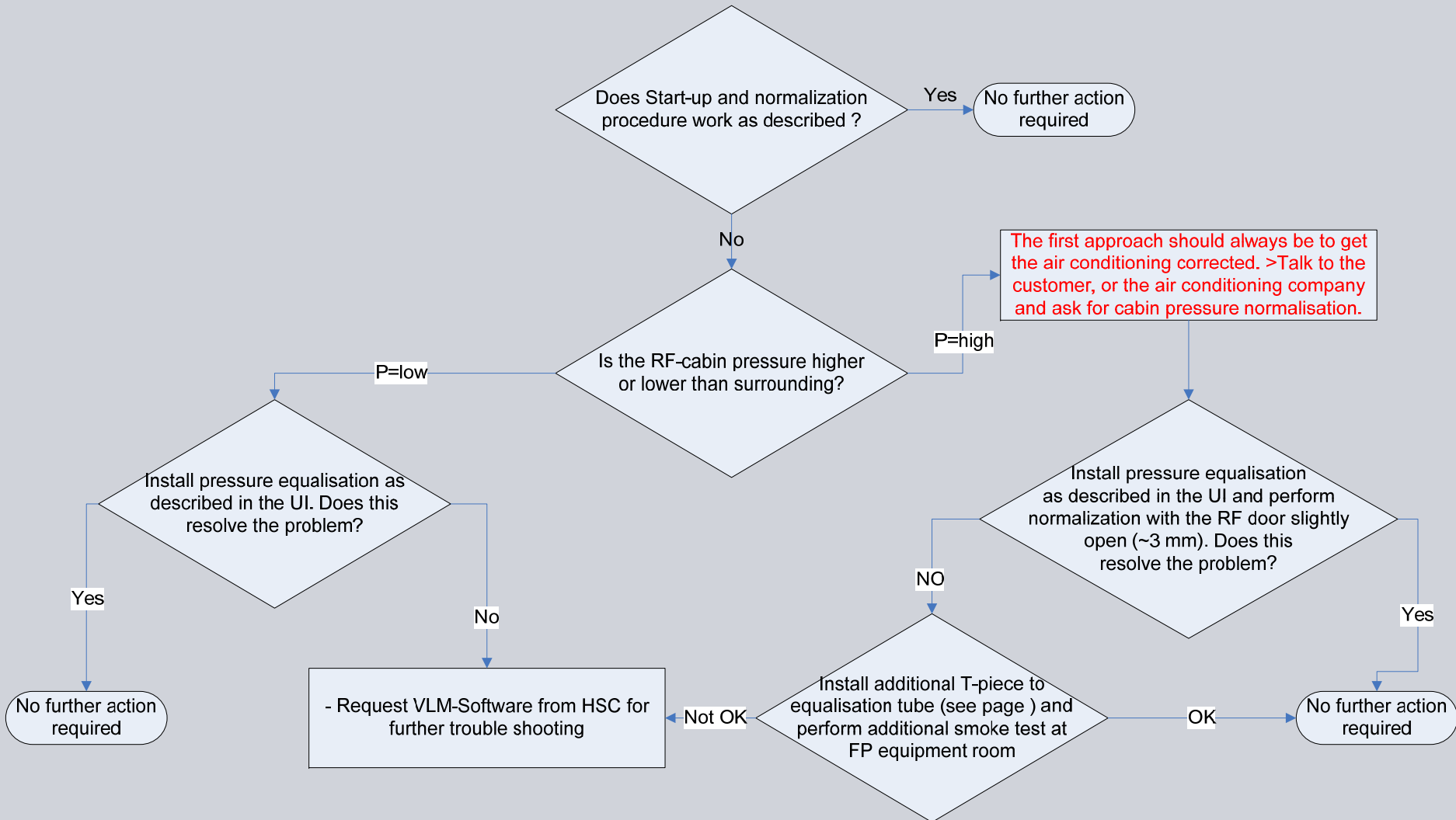


4. Missing sealing in joint elements.
In case of a FAULT condition, check the joint elements (T-pieces) for proper sealing. There must be a white rubber seal as shown on the left in each joint. A missing seal results easily in an air flow difference of 1.4 ltrs./min. (~7%).



5. The black bushing must always be used when a hose is being connected to the inlet or outlet of the VLM. If missing, the flow will increase dramatically for 9 ltrs./min. (~43%). When the grey outlet cover is being removed, the flow increases for 0.4 ltrs./min. (~2%)

How to proceed, when air-pressure related problems occur during VLM start-up



Installation of an additional T-piece in the pressure equalisation hose



Perform a normalisation with the RF cabin door slightly open (~3 mm) first. This results in a normalisation at a mean cabin air pressure between maximum (door closed) and minimum (door open) and gives you a maximum of tolerance from normalised value in both directions.

If this does not resolve the problem, an additional T-piece looped into the pressure equalisation hose may help. Cut the pressure equalisation hose about 1 m behind the VLM exhaust (see left pic.)

Installation of an additional T-piece in the pressure equalisation hose



Prepare a T-piece with appr. 2 m of hose as shown on the left.

The T-piece can be ordered as a standard spare part with part no.: 10683922 (this T-piece has an additional reduction to 10 mm fitted, which must be removed).

If needed, the black bushing at the VLM-in/outlet can be also ordered as a spare part: 10683929.

Connect the T-piece in line of the previously cut pressure equalisation hose as shown on the left.

> The outlet of the 2 m hose must remain inside the equipment room.



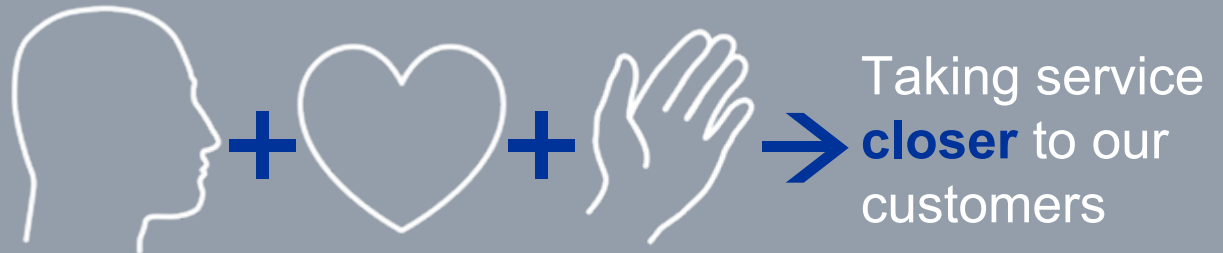
Installation of an additional T-piece in the pressure equalisation hose



After the T-piece has been installed, an additional smoke test must be performed at the gradient filter cover sample point inside the equipment room, in order to verify, that there is still being air sucked into the VLM device.

Remove the gradient filter cover from the filter panel in the equipment room and perform a smoke test as described in the UI.

> If the smoke test does not work, advanced trouble shooting is necessary and the VLM-software should be requested via HSC.



Think. Feel. Do. **Make the difference.**

Contact

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